

ITMO UNIVERSITY

CERTIFICATE

This is to certify that

Yagyaansh Khaneja

born on 23.11.1998,

has taken part in the Research Summer Camp in Photonics

at ITMO University,

3 ECTS credits

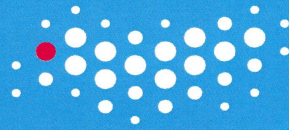
held on May 27 – June 7, 2019

First Vice Rector



Daria K. Kozlova

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**The following topics and modules
have been covered within the program:**

- Introduction in the field of quantum information and quantum calculation
- Introduction to the terahertz optics
- End-to-end optimized camera imaging
- Review of Spatio-temporal holography
- Workshop of the Laboratory of digital and display holography
- Quantum communications
- Application of femtosecond lasers in medicine and biology
- Introduction to Digital holographic microscopy and tomography
- Optical nonlinearities and ultrafast dynamics in media: experimental techniques. Z-scan method and pump-probe spectroscopy
- Optical tomography for biomedical application
- Application of terahertz optics for biomedical use, food and art-works testing and bio-like materials
- Single photon sources and detectors
- Security of QKD systems
- Workshop «Wavefront propagation»
- Research topic:
 - THz generation from laser filamentation in the different liquids

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